

Madhavan Balasubramaniam

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PROFILE SUMMARY

MSCS student at Georgia Tech with 3+ years of experience as a Machine Learning Engineer at Bosch, specializing in Python, Classic ML NLP, GenAI and deploying ML systems. Experienced in building and automating backend and cloud-based (Azure) ML solutions.

EDUCATION

Georgia Institute of Technology, Atlanta, GA

May 2027 (Expected)

Masters of Science in Computer Science - Machine Learning Specialization

Relevant Coursework: *Machine Learning for Trading, Bayesian Statistics, Machine Learning, Deep Learning*

National Institute of Technology Trichy

May 2022

Bachelors in Instrumentation & Control Engineering (minor in Computer Science)

GPA: 3.81/4.0 Rank: 2/10

Relevant Coursework: *Data structures, Databases, Operating Systems, Networking and Protocols*

WORK EXPERIENCE

Bosch Digital Twin - ADNOC (ML, Datascience)

July 2024 - Dec 2025

- Developed and delivered a multimodal ML + physics-based predictive maintenance system, diagnosing 100+ faults (e.g., mechanical seal failure, impeller thinning, clearance faults) across 20+ turbomachineries.
- Optimized ML and time series forecasting models (Prophet (meta), XGBoost, Polynomial Regression) for Remaining Useful Life prediction and fault severity classification of machine components.
- Leveraged MLflow for experiment tracking, model version control, and performance benchmarking, enabling reproducible and improved prediction accuracy through hyperparameter optimization.
- Implemented operational KPI such as MTTF, MTBF, MRR for dashboard enabling proactive maintenance and minimizing downtime.

Bosch Automation Platform (Devops, NLP)

August 2022 - July 2024

- Optimized metrics dashboard by restructuring data storage and implementing indexed aggregation Mongo queries, improving loading speed by 10x. The dashboard provided insights into platform usage, directly contributing to a **USD 100K contract renewal**. Recognized with the **Stand-out Performer Award** for this achievement.
- Developed Python-based chatbot (microservices) backend using FastAPI, supporting async CRUD operations and file handling via Nginx and RabbitMQ, with MongoDB queries optimized for pagination on frontend.
- Deployed services and CronJobs on Red Hat OpenShift by creating the deployment, service and ingress config files and automated using Ansible file (CI/CD)
- Designed hybrid document retrieval system: semantic search (RoBERTa embeddings + cosine similarity) with keyword search (BM25) using Elasticsearch as a vector store for BAP (Document base) platform.
- Fine-tuned (SFT) BART Summarizer (LLM)** with custom training data and deployed it using Docker thereby enriching the user experience for semantic search on the BAP (Document base) platform.

PROJECTS

UNVotePredictor (SocWebLab, GaTech - Supervisor: Prof. De Choudhury)

Jan 2026 - Present

- Architected UNVotePredictorAttention, a cross-attention transformer that encodes UN resolution text via ModernBERT (8,192 tokens) and attends over 193 learnable country embeddings to jointly predict Yes/No/Abstain votes for all member states in a single forward pass.
- Trained on 15K resolution text and 947K voting records (1945–2019) on 8× L40S GPUs with DataParallel, gradient accumulation, and early stopping on masked cross-entropy loss, achieving 87.75% weighted F1 on a temporally split held out test set.

DeepDoc.AI [Demo]

May 2025 - July 2025

- Developed a Python-based RAG chatbot for intelligent document analysis using LangChain and Ollama's Llama 3.2 model, enabling contextual question answering across multiple documents.
- Implemented a hybrid retrieval pipeline combining BGE embeddings, BM25, and FAISS with recursive chunking and adaptive sub-query generation, improving retrieval accuracy and document insight extraction.
- Built multimodal parsing and cross-document analysis capabilities for text, tables, and images, integrated conversation memory for contextual continuity, and applied prompt engineering for structured output generation.

SKILLS

Programming : Python, SQL, Unix Bash, MATLAB, C/C++, markup - HTML & CSS

ML Libraries & Frameworks: PyTorch, LangChain, LangGraph, scikit-learn

Databases: MongoDB, MySQL, Redis, Elasticsearch, FAISS

DevOps & Cloud: Docker, Red Hat OpenShift, Ansible, Airflow, MLflow, SonarQube, Git, Azure ML, Azure AKS